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AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

Faculty of Science & Technology

Department of Computer Science

CSC 3224: Computer Graphics (All Sections)

Set-A

Mid-Term Examination Semester: Summer 2023-24

Total Marks: 100 Moderator: Aneem Al Ahsan Rupai Time: 2 hours

Instructions:

1) There are **4** Parts **[Part-A, Part-B, Part-C and Part-D]**. Answer from all the Parts.
2) Answer **Part-A** (MCQ and True/False) in the **OMR** and other parts in the spaces provided in this question.
3) **Part-B** is for **OBE** evaluation. Answering this part is **MANDATORY**. Answer in the provided space inside the question paper.
4) Use of calculator is **ALLOWED** during examination.
5) [redacted]

Nam[redacted]	**Proctors Sign:**
ID[redacted]	
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Par[redacted]

Answer this part in OMR. Darken the correct answer in the OMR. Make sure you darken your ID properly. **30 marks**

Multiple Choice Question

1 Halftone breaks up an image into a series of

A. Pixel B. Color C.Shade D.None

2 Which one is not a format of Raster graphics?

A. TIFF B. PCX C. PICT D. GIF

3 In coloring, the technique that uses a storage approach to identify the color codes is:

A. Direct Coding B. Lookup Table C. Rasterization D. Halftone

4 What is the formula for the decision parameter in East for Bresenham's midpoint circle algorithm:

A. $d_0 + 2x + 3$ B. $d_0 + 2(x - y) + 5$ C. $1 - R$ D. None

5 If point A (2, 3) belongs to a line having m = 2.5, what will be the next point B if DDA is applied?

A. B(3, 3) B. B(2, 4) C. B(3, 4) D. B(4, 4)

6 If d >= 0, then next which point is selected according to midpoint circle algorithm?

A. East B. North-East C. South-East D. Both a & b

7 Which one is a callback function?

A. glColor3ub() B. glEnd() C. glutDisplayFunc() D. glFlush()

8 For the equation $(x+5)^2 + (y-7)^2 = 16$ The initial Y co-ordinate for Bresenham's midpoint line algorithm is generally considered:

A. 16 B. 4 C. -5 D. 7

9 More than 4 vertices in a primitive means:

A. Line B. Quad C. Polygon D. None

10 An image composed of individual pixels arranged in a grid is referred to as:

A. Vector graphic | B. Raster image | C. Polygon mesh | D. Bezier curve

11 For a vector image the mathematical formula is used-

A. To find the pixels | B. To find the color shading | C. To find the coordinates that are mapped to pixels | D. None

12 For a raster image pixel can be arranged in a regular

A. 1D Grid | B. 2D Vector | C. Bitmap | D. Coordinates

13 $(x-1)^2 + (y-1)^2 = 25$, considering the equation, which one of the following would be the initial point according to Bresenham's midpoint circle algorithm?

A. (0, 25) | B. (0, 5) | C. (1, 1) | D. (-1, -1)

14 If $d > 0$, then which one is true for midpoint line algorithm?

A. $d_{start} = 2d_y - d_x$ | B. East point is selected if $d > 0$ | C. $(\Delta d)_E = 2d_y$ | D. $(\Delta d)_{NE} = 2(d_x - d_y)$

15 Which of the following is a coloring technique?

A. Direct Coding | B. Lookup Table | C. Both A and B | D. None

True/False
For 'True' darken A and for 'False' darken B in the OMR

16 Raster Image is better for designing a logo

17 To create polygon primitive at most 3 points are needed.

18 Raster images use mathematical calculations.

19 Additive color model uses CMY color space.

20 Raster images use mathematical calculations

Part-B	15 marks
CO2 Description: Analyze the [redacted] able to [redacted]	
N[redacted]	

14

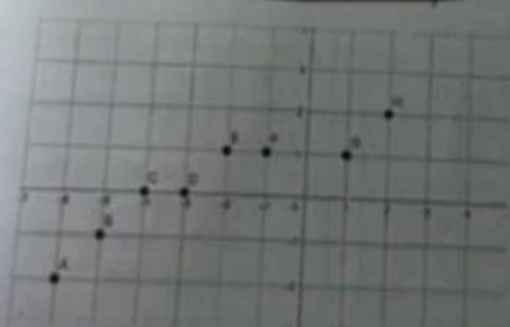

Figure: Points generated after applying a line drawing algorithm

...ordinates shown in the diagram shown above were generated after applying a line drawing ...ithm. Answer the following-

a. Identify the algorithm that was applied and give possible explanations for obtaining the coordinates as shown in the diagram

The algorithm was ~~dt~~ DDA line drawing algorithm. It is used to create line using round function. The coordinates are obtained by round function, hence the coordinates were obtained.

b. Discuss what type of problem you will encounter when you use the same algorithm to draw a line with coordinates A(2,5) and B(2,9).

If we use DDA on A(2,5) and B(2,9):

$m = \frac{\Delta y}{\Delta x} = \frac{9-5}{2-2} = \frac{4}{0}$ which is not calculatable. As a result we can't use the value of m to draw the line and without the m value, line ~~dr~~ drawing is not possible in DDA.

c. Discuss another algorithm that might solve the positioning problem of the algorithms and write down the formulas for that algorithm.

Bresenham's midpoint line algorithm might solve the problem. The formulas for it are:

When $m<1$:

$d_{start} = 2d_x - d_y$

$(\Delta d)_{NE} = 2(d_x - d_y)$

$(\Delta d)_E = 2d_x$

$d \leq 0 \rightarrow E\ (x_0+1, y_0)$

$d > 0 \rightarrow NE\ (x_0+1, y_0+1)$

When $m>1$:

$d_{start} = 2d_y - d_x$

$(\Delta d)_{NE} = 2(d_y - d_x)$

$(\Delta d)_N = 2d_y$

$d \leq 0 \rightarrow N\ (x_0, y_0+1)$

$d > 0 \rightarrow NE\ (x_0+1, y_0+1)$

It only uses arithmetic functions and no rounding functions. It might solve the positioning this way.

Part-C Derivation	10 marks

1. Show the derivations for the formulas that are required to draw a circle using Bresenham's Midpoint Circle Algorithm.

If the equation is $x^2+y^2=R^2$

The initial point will be $(0,R)$

how?

Decision parameter:

$M = (x+1, y+\frac{1}{2})$

$d_{start} = F(M) = 1^2 + \frac{1}{4}R^2 + R^2$

$= \frac{5}{4} - R = 1 - 4(\frac{5}{4} \times 1)$

how?

For East:

$M_E = (x_0+2, y_0+\frac{1}{2})$

$d_E = \cancel{d_0} F(M_E)$

$= (x_0+2)^2 + (y_0+\frac{1}{2})^2 - R^2$

$= x_0^2+2x_0+4+y_0^2+\frac{1}{2}y_0+\frac{1}{4}-R^2$

$= (x_0+1)^2 + (y_0+\frac{1}{2})^2 - R^2 + 2x_0+3$

$= d_0+2x_0+3$

4

For d_{SE}: $M_{SE} = (x_0+2, y_0+\frac{3}{2})$

$d_{SE} = F(M_{SE})$

$= d_0 + 2(x_0-y_0)+5$.

7

Part-D Answer any 3 out of the 4 questions	15X3=45 marks

1. (a) Suppose you have 2 coordinates as initial points A (1,2) and D (4,4). As part of scan conversion using Midpoint Line Algorithm, you obtained two more points B (2,3) and C (3,3) in between the initial and final points. What was the decision taken (East or North-East) to reach C from B. Write down the formula used for this decision and show the derivation for this formula.

$m = \frac{y_n-y_0}{x_n-x_0} = \frac{4-2}{4-1} = \frac{2}{3} = \frac{2}{3} = 0.667 < 1$

3

From B to C,

$\Delta y = 3-3=0$ and $\Delta x = 3-2 = 1$

Here, x value incremented, so the decision was East

The formula is:

$(\Delta d)_E = 2d_x$

The derivation is

$d_{new} - d_{old} = f(x_0+1, y_0) - f(x_0, y_0)$

$= a(x_0+1) + by_0 + B - (a(x_0) + by_0 + B)$

$= ax_0 + a + by_0 + B - ax_0 - by_0 - B$

$= a = d_y = 2d_x$

$(\Delta d)_E = 2d_x$

(b) For the given equation of a circle $(x+2)^2 + (x-3)^2 = 49$

i. Find the center of the circle

ii. Find the radius of the circle

iii. Initial coordinate

(i) Center is $(-2, 3)$

(ii) Radius is, $R = \sqrt{49} = 7$

(iii) Initial coordinate $= (0, R) = (0, 7)$

5

2. (a) For a given starting pixel of a straight-line A (3,5) and ending pixel B(22, 32) show the first five-iterations using DDA Algorithm.

$m = \frac{\Delta y}{\Delta x} = \frac{32-5}{22-3} = \frac{27}{19} = 1.42 > 1$

Formula: $x_{new} = (x_{old} + \frac{1}{m})$, $y_{new} = \cancel{round(y_{old} + \frac{1}{m})}\ y_{old} + 1$

x_{old}	y_{old}	x_{new}	y_{new}	x_p	y_p
3	5	$3 + \frac{1}{1.42} = 3 + 0.7 = 3.7$	$5 + 1 = 6$	4	6
3.7	6	$3.7 + 0.7 = 4.4$	$6 + 1 = 7$	4	7
4.4	7	$4.4 + 0.7 = 5.1$	$7 + 1 = 8$	5	8
5.1	8	$5.1 + 0.7 = 5.8$	$8 + 1 = 9$	6	9
5.8	9	$\cancel{6.5}\ 5.8 + 0.7 = 6.5$	$9 + 1 = 10$	7	10

10

(b) Convert the following as per instructions.

i. Convert CMY color with code (11%, 36%, 77%) into RGB color code.

ii. Convert RGB color with code (12, 125, 77) into CMY color code.

(i) $R = (1 - \frac{11}{100}) * 255 = 226.95$

$G = (1 - \frac{36}{100}) * 255 = 163.2$

$B = (1 - \frac{77}{100}) * 255 = 58.65$

$\therefore RGB = (226.95, 163.2, 58.65)$

(ii) $C = 1 - \frac{R}{255} = 1 - \frac{12}{255} \times 100\% = 95.29\%$

$M = 1 - \frac{G}{255} = 1 - \frac{125}{255} \times 100\% = 50.98\%$

$Y = 1 - \frac{B}{255} = 1 - \frac{77}{255} \times 100\% = 69.80\%$

$\therefore CMY = (95.29\%, 50.98\%, 69.80\%)$

3. (a) Define the following terms related to Computer Graphics:

i. Direct Coding
ii. Lookup Table
iii. Colour Model
iv. Pixel
v. Rasterization

(b) Explain the steps required in Scan Conversion

(c) Differentiate between Raster and Vector images.

4. (a) Show mathematically how you can create different color palettes each containing different number of color shade.

3 bit RGB:

If we assign one bit per color in RGB, we get an 8 color $((2^1)^3)$ color pallete.

6 bit RGB:

If we assign 2 bits per color in RGB, we get $((2^2)^3) = 64$ ~~bit~~ color color pallete.

(b) Differentiate between Direct Coding and Lookup Table

Direct coding uses storage for each pixel to color code it. In Lookup table, a table is already setup with color to take and color pixel using it.

(c) What are the color models and their corresponding color spaces?

The color models are RGB and CMYK. The color spaces are RGB for Red, Green and Blue and for CMYK Cyan, Magenta, Yellow, Black.